ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING (EEE)

COURSE: EEE 4484 (DIGITAL ELECTRONICS & PULSE TECHNIQUES LAB.)
EXPERIMENT NO. 03

STUDY OF OPERATIONAL AMPLIFIER AS TRIANGULAR WAVE GENERATOR & ASTABLE MULTIVIBRATOR

OBJECTIVE:

1. To study the basic use of an Operational Amplifier (Op-Amp).
2. Study of bipolar triangle wave generator using Op-Amp.
3. Study of unipolar triangle wave generator using Op-Amp.
4. To build Astable Multivibrator using Op-Amp.

INTRODUCTION:

Operational Amplifier (Op-Amp) can be utilized to many electronic circuits. Bipolar triangle wave generator and unipolar triangle wave generator can be simply made up with Op-Amps. Also, in many applications, designer requires astable multivibrators mainly in digital circuits. Op-amp with diodes and a few resistors in the circuit can be built to realize the above-mentioned circuit performances.

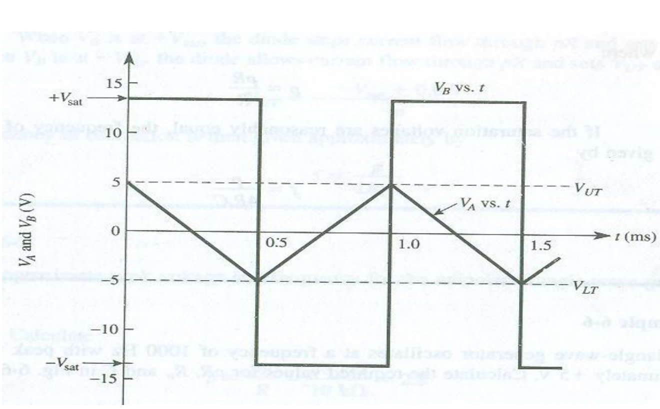


Fig. 3(a): Waveshapes of Bipolar Triangle Wave Generator

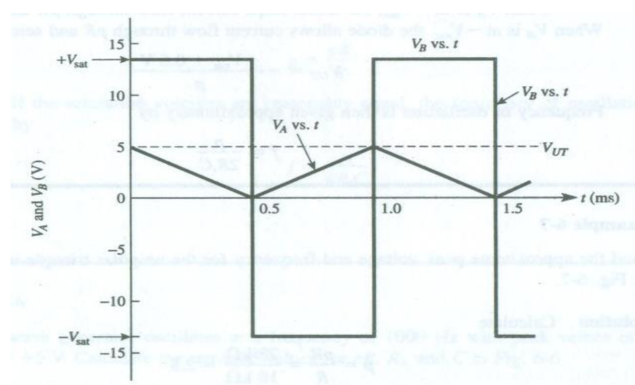


Fig. 3(b): Waveshapes of Unipolar Triangle Wave Generator

Astable multivibrator is an electronic device which can continuously shift between its two states with respect to the output. If the output corresponding to one particular state is high, then the output corresponding to the other state is low. This nature of the circuit is useful in producing continuous output stream of pulses. There is no need for external triggering in case of astable multivibrator circuits. Simply turn on the power and the circuit start oscillating between its two states. Since it is an oscillator, a positive feedback can be found in this kind of circuits also.

EQUIPMENT:

1. DC Power Supply,
2. Function/Signal Generator,
3. Breadboard,
4. Resistors,
5. Diode (IN 4007),
6. Capacitors,
7. Op-Amp (LM 741),
8. Oscilloscope, and
9. Connecting wires.

KNOWLEDGE PROFILE AND APPLICATION:

1. Basic Working Principle of Operational Amplifier (Op-Amp),
2. Bipolar & Unipolar Triangle Wave Generator using Op-Amp, and
3. Configuration & Working Principle of Astable Multivibrator.

PROCEDURE:

1. Connect the following circuits given in Fig. 3(c), Fig. 3(d) and Fig. 3(e) in breadboard.
2. Observe the waveshapes in Oscilloscope & plot all the graphs on a single graph sheet as shown beside each Fig. 3(c), Fig. 3(d) and Fig. 3(e).

CIRCUIT DIAGRAM:

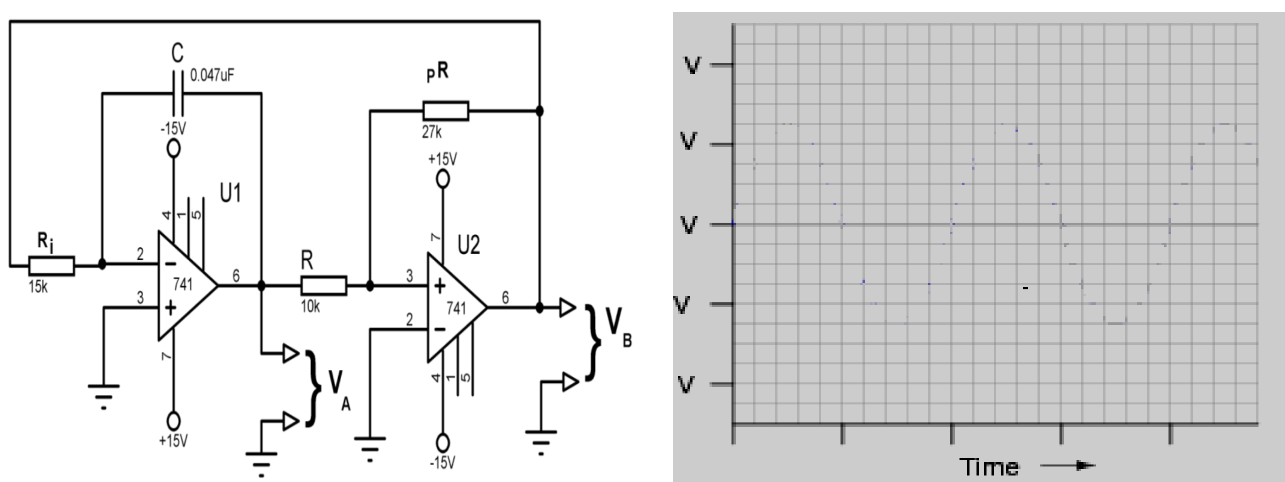


Fig. 3(c): Bipolar triangle wave generator

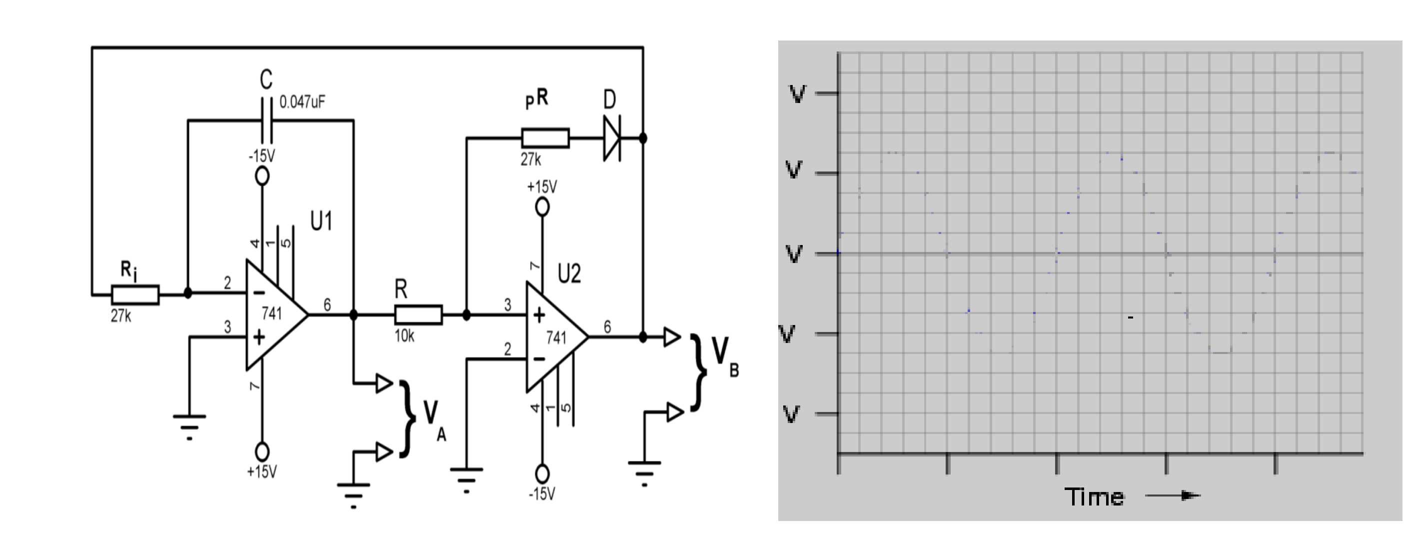


Fig. 3(d): Unipolar triangle wave generator

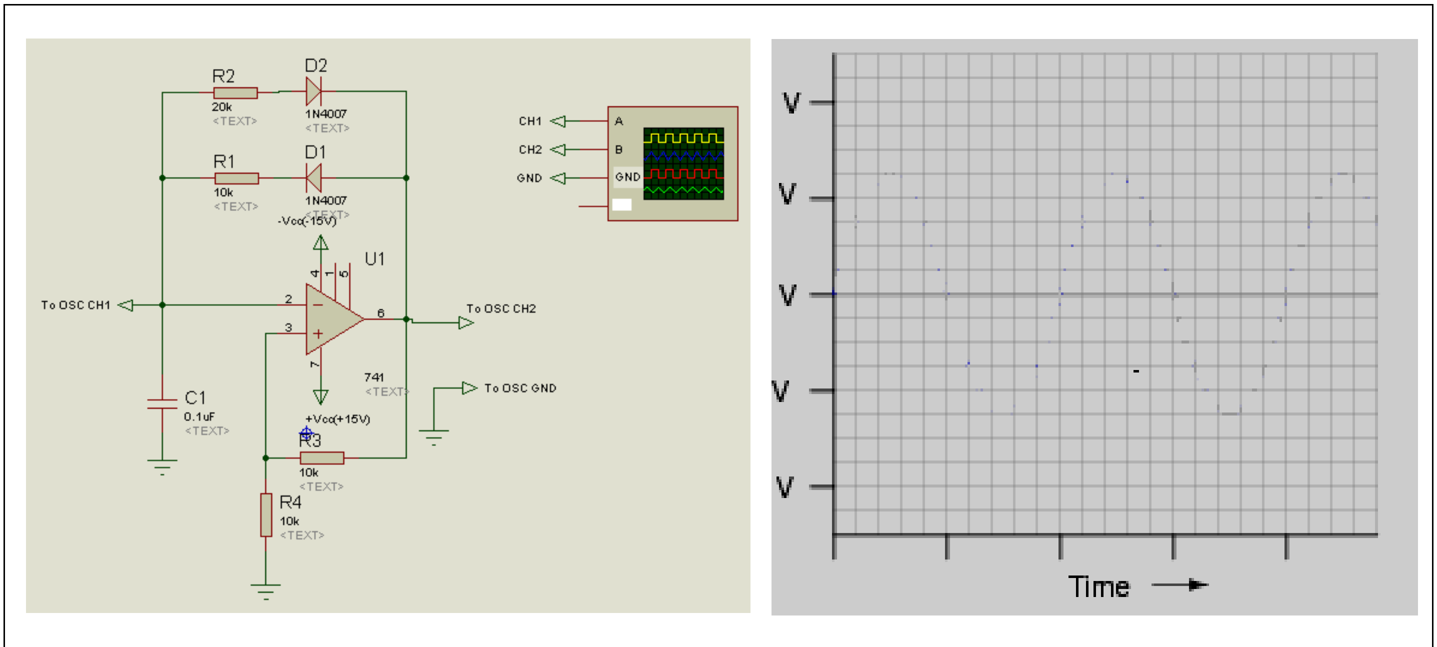
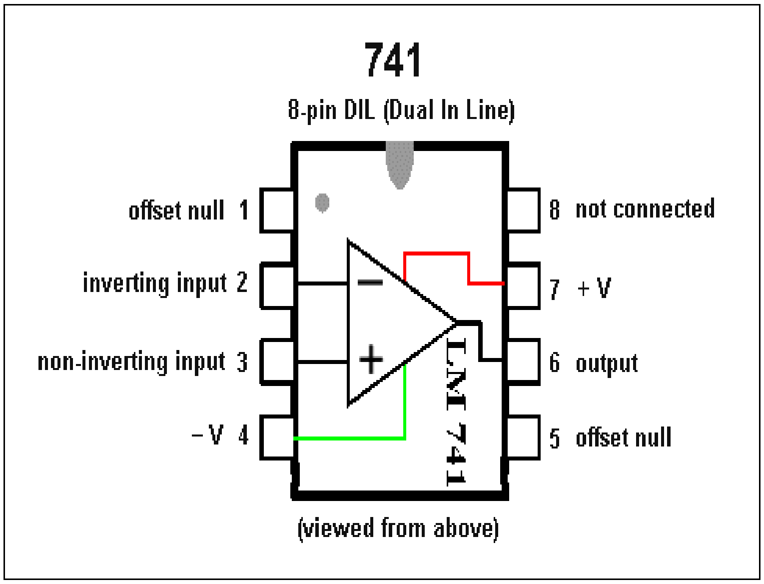


Fig. 3(e): Astable Multivibrator

PIN CONFIGURATION OF OP-AMPS:



REPORT QUESTION:

1. Analyze the differences between the circuits of Fig. 3(c) and Fig. 3(d).
2. Describe the circuit of Fig. 3(e) on the basis of its' working principle.
3. Write a short discussion on the conducted experiment.